

CEH v13 Practice Questions – Module 20: Cryptography

1. Which algorithm is considered symmetric encryption?

- A. RSA
- B. AES
- C. ECC
- D. DSA

Answer: B

2. What does a digital signature provide?

- A. Confidentiality
- B. Non-repudiation
- C. Obfuscation
- D. Compression

Answer: B

3. Which encryption method uses two different keys for encryption and decryption?

- A. Symmetric encryption
- B. Asymmetric encryption
- C. Stream cipher
- D. Block cipher

Answer: B

4. What is the key size of AES-256 encryption?

- A. 128 bits
- B. 192 bits
- C. 256 bits
- D. 512 bits

Answer: C

5. Which of the following is a hashing algorithm?

- A. DES
- B. RSA
- C. MD5
- D. Blowfish

Answer: C

6. Which of the following is a characteristic of a hash function?

- A. It is reversible
- B. It produces a variable-length output
- C. It provides one-way encryption
- D. It encrypts data for storage

Answer: C

7. What is the purpose of salting passwords before hashing?

- A. To increase speed
- B. To prevent brute force
- C. To defeat rainbow table attacks
- D. To reduce hash length

Answer: C

8. Which standard is used for public key certificates?

- A. SHA-1
- B. AES-256
- C. X.509
- D. DES

Answer: C

9. Which protocol uses asymmetric encryption during the handshake and symmetric thereafter?

- A. FTP
- B. SSH
- C. TLS
- D. SNMP

Answer: C

10. What does PKI stand for?

- A. Public Key Infrastructure
- B. Private Key Identifier
- C. Public Key Integrator
- D. Password Key Interface

Answer: A

11. What is the primary weakness of the MD5 algorithm?

- A. Long key sizes
- B. Poor performance
- C. Vulnerability to collisions
- D. Lack of availability

Answer: C

12. Which algorithm is primarily used in blockchain technology?

- A. RC4
- B. SHA-256
- C. DES
- D. 3DES

Answer: B

13. Which of the following uses a public and private key pair?

- A. AES
- B. Blowfish
- C. RSA
- D. DES

Answer: C

14. Which cryptographic function transforms plaintext into ciphertext?

- A. Encoding
- B. Hashing

- C. Encryption
- D. Compression

Answer: C

15. Which encryption technique splits data into fixed-size blocks?

- A. Stream cipher
- B. Block cipher
- C. Asymmetric cipher
- D. Substitution cipher

Answer: B

16. What is the role of a certificate authority (CA) in PKI?

- A. Create private keys
- B. Issue and manage digital certificates
- C. Encrypt user data
- D. Manage session tokens

Answer: B

17. Which of the following is a common use of cryptography?

- A. Antivirus scanning
- B. Packet routing
- C. Secure communication
- D. Network discovery

Answer: C

18. Which tool is commonly used to inspect SSL/TLS certificates?

- A. Netcat
- B. OpenSSL
- C. Nmap
- D. Burp Suite

Answer: B

19. Which type of attack attempts to find two inputs that produce the same hash?

- A. Brute force
- B. Collision attack
- C. Man-in-the-middle
- D. Replay attack

Answer: B

20. What is the key size for RSA commonly used in secure communication?

- A. 128 bits
- B. 256 bits
- C. 1024 bits
- D. 2048 bits

Answer: D

21. What cryptographic concept ensures a message has not been altered?

- A. Confidentiality
- B. Authentication
- C. Integrity
- D. Availability

Answer: C

22. Which term describes encrypted data?

- A. Plaintext
- B. Ciphertext
- C. Metadata
- D. Signature

Answer: B

23. What is steganography?

- A. A form of symmetric encryption
- B. Hiding data within other media files
- C. A type of hashing algorithm
- D. Encoding passwords

Answer: B

24. Which function ensures non-repudiation in digital communication?

- A. Encryption
- B. Hashing
- C. Digital signature
- D. Encoding

Answer: C

25. Which cryptographic attack involves trying every possible key?

- A. Replay attack
- B. Man-in-the-middle
- C. Brute-force attack
- D. Dictionary attack

Answer: C